Distributed Consensus Protocols

Raft and Paxos guarantee linearizable state machine replication across server quorum

Structural verification mechanisms for fault-tolerant computing.

Database Index Structures

B-trees provide predictable logarithmic search times while LSM trees minimize write ar

Structural verification mechanisms for fault-tolerant computing.

Event Streaming Pipelines

Partitioned commit logs decouple high-throughput producers from decoupled downstre

Structural verification mechanisms for fault-tolerant computing.

Telemetry and Observability Infrastructure

Comprehensive observability requires unified distributed tracing, metric counters, and centralized structured log aggregation. Standards like OpenTelemetry correlate span contexts across microservice boundaries to rapidly detect latency regressions.